Brad Cao

(301) 356-1069 | bcao4049@gmail.com | github.com/bradca0

Education

Columbia University New York, NY
B.A. in Mathematics & Computer Science *Expected May 2029*

Relevant Coursework: Data Structures in Java, Probability Theory, Natural Language Processing, Honors Mathematics A & B

Activities: Columbia Alpha Partners (Global Markets Division), Columbia Artificial Intelligence Alignment Technical Fellowship, Columbia Poker Club

Experience

Research Fellow, SPAR (Supervised Program for Alignment Research) · *Remote* Sep. 2026 – Present
- Build evaluation datasets that test whether humans working with AI "judges" can reliably oversee other AI systems, as part of a 100+ dataset leaderboard covering deception, jailbreak robustness, cultural bias, and computer-control mistakes
- Audit existing judge datasets for coverage gaps and create new ones to fill them, using sandwiching for ground truth
- Measure how easily judges are fooled by reward hacking under adversarial RL optimization

Data Science Intern, Mercor · *San Francisco, CA* Apr. 2026 – Sep. 2026
- Developed 10+ rubric-based scoring taxonomies that turn expert judgment on professional tasks into consistent criteria for evaluating and training LLMs (RLHF annotation schemas)
- Drafted pass/fail criteria with acceptable and unacceptable example responses so expert reviewers score consistently
- Flagged edge cases (ambiguous, incomplete, or conflicting inputs) that ordinary prompts miss and added them to the rubrics
- Audited labeled data for reviewer disagreements and suggested guideline updates to raise inter-annotator agreement

Research Assistant, Duke University · *Durham, NC* Apr. 2026 – Sep. 2026
- Designed and built an LLM pipeline (GPT-5.5, OpenAI Batch API) that analyzed 18.7M words across 3,543 cable news transcripts (CNN, Fox, MSNBC; 2011–2025), one batch job per transcript
- Refined a 20-page coding protocol over 18 versions to reach ∼99.4% accuracy on flagged cases with no missed cases; cleaned the data
- Built an auditable review workflow: 12,892 flagged candidates, 1,584 ambiguous (12.3%) sent to human review; found Fox disparages media at 10× CNN's rate, 4× MSNBC's

Leadership Seminar Scholar, University of Notre Dame · *South Bend, IN* Jun. 2025 – Aug. 2025
- Won a full-ride scholarship to Notre Dame's "Inequality in America" seminar; conducted original meta-research on how the pharmaceutical industry shapes health policy, reviewing 15+ studies
- Ran regression analysis and visualizations in R linking industry practices to poverty outcomes, connecting to the seminar's focus on economic inequality; presented at a symposium

Research Intern, University of Maryland, Baltimore · *Baltimore, MD* Jun. 2023 – Sep. 2025
- First-authored 2 studies, presented at the ASCPT Annual Meetings (2024, 2025); the 2025 poster won the Top Poster Award
- Built ML models (logistic regression, random forests) on 10 years of FDA adverse-event reports to predict serious outcomes in ADHD patients on stimulants; tuned with cross-validation; best model reached 0.75 AUC and 88% recall
- Modeled tacrolimus dose requirements and transplant outcomes in lung transplant recipients taking azole antifungals

Projects

Threshold | *Python, NumPy, SciPy, PyTorch* | github.com/bradca0/threshold Solo project
- Reproduced double descent (test error that worsens, then improves, as a model grows) on MNIST with random-feature regression (2,000 samples, 20% label noise): error peaks at 88.8% when parameters = samples, vs. 24.3% at 5× more
- Identified the cause: the solution's norm explodes at the peak (811 vs. 2.6) and 90% of the peak error is variance; tuned ridge regularization cut the peak from 88.8% to 12.3%
- Showed more data can hurt (error 34.8% to 89.1% as samples grew 250 to 2,000); replicated on Fashion-MNIST; an MLP showed no peak (reported as a negative result); wrote 85 tests at 94% coverage

MyCureHope | *Django, PostgreSQL, JavaScript, TailwindCSS* | mycurehope.org Co-developer
- Co-developed an open-source database of treatments, doctors, and hospitals used by 33,000+ users across 78 countries
- Personally built the search feature, own the technical development, and deployed the full stack on Nginx/Gunicorn/Ubuntu

Publications

Cao, B., Lee, S., Naga, A., Gobburu, J., Dunn, A. "Leveraging Supervised Machine Learning to Identify Predictors of Serious Clinical Outcome with Stimulant Treatment from Real-World Data." *ASCPT Annual Meeting*, 2025.

Cao, B., Gupta, P., Dunn, A. "Predictive Modeling of Tacrolimus Dose Requirements and Transplantation Outcomes in Lung Transplant Recipients on Systemic Azole Antifungals." *ASCPT Annual Meeting*, 2024.

Technical Skills & Interests

Languages: Python, JavaScript, Java, SQL (PostgreSQL), R, HTML/CSS
Tools & ML: PyTorch, TransformerLens, NumPy, pandas, scikit-learn, Django, Linux; mech interp, LLM fine-tuning, RLHF
Interests: Fencing (USA Fencing First Team All-American), Competitive Programming (USACO), History Olympiads, Open-Source